

Dani R. Lipman

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RESEARCH FOCUS

I am interested in exploring dynamics and 3-D modeling of galaxy centers using both observations and MHD simulations. I am particularly interested in understanding how material may fall off nuclear orbits and inflow closer to circum-nuclear discs near central supermassive black holes. My main question is: How do dynamics and physical parameters work together to control star formation and galaxy evolution?

EDUCATION

- **University of Connecticut** August 2020 - Present
Physics Ph.D Candidate Storrs, CT
 - Expected graduation May 2025
 - MSc in Physics conferred May 2023
- **University of Iowa** 2015 - 2019
Undergraduate Education Iowa City, IA
 - B.S. in Physics and Astronomy (Double Major), Mathematics Minor
 - Received University Honors; Honors in Astronomy

HONORS AND AWARDS

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| 2024 | NASA Future Investigators in NASA Earth and Space Science and Technology (FINESST) Fellow
Title: <i>Central Station: Mass Transport into the Galactic Center with AREPO</i> |
| 2022 | Connecticut Space Grant Consortium Graduate Research Fellowship
Title: <i>3D CMZ: Distinguishing Near vs. Far Distances in the Galactic Center Using Spitzer and Herschel</i> |
| 2019 | China Fulbright Study/Research Award
Title: <i>Investigating Perception and Design of Science Outreach in Chinese Universities</i> |
| 2019 | UIowa Physics Department Distinguished Service Award |
| 2017 | University of Iowa Latham Science Engagement Initiative Fellow
<i>Yearlong course to develop science outreach programs. Developed group and individual outreach projects aimed at high school and general audiences, respectively.</i> |
| 2017; 2018 | Iowa Space Grant Consortium Undergraduate Research Fellowship
Title: <i>Characterizing the Structure of the Martian Bow Shock</i> |

RESEARCH EXPERIENCE

Graduate Research

- **3D Central Molecular Zone** 2020 - Present
3D CMZ Paper Series: https://centralmolecularzone.github.io/3D_CMZ/
 - Developed three new dust extinction techniques to differentiate near/far positions of molecular clouds in the Galactic Center.

- Created a Bayesian framework to synthesize existing datasets for Near/Far position distinctions in the CMZ.
- Developed 4-D (ℓ, b, v and near/far) fitting method of an x2 elliptical orbit for the Milky Way's CMZ.
- **Development of MHD Zoom-in Simulations** 2022 - Present
 - Leading the development of solar mass resolution zoom-in Simulations of Milky-Way like galaxies using the moving mesh code AREPO. Simulations combine updated external gravitational potential from Hunter et al. (2024), magnetic fields, star particles [representing clusters], accreting sink particles [representing prestellar cores], and supernova feedback.
- **Identification of High-Velocity Features in the CMZ with ACES** 2025 - Present
 ALMA CMZ Exploration Survey (ACES): <https://sites.google.com/view/aces-cmz/home>
 - Led catalog creation for high-velocity compact clouds seen in high resolution ALMA CS(2-1) maps of the CMZ. The catalog provides a list of potential inflow candidates from the CMZ ring towards the circumnuclear disk surrounding SgrA*.
 - Led intensive 2-week paper sprint to perform analysis and produce a full paper draft with a small group within 2-weeks.
- **China Fulbright Study/Research Fellow** 2019 - 2020
 Nanjing University, China
 - Independent research designed to investigate the perceptions and methods of university faculty at Chinese universities, particularly in Astronomy, participating in public outreach.
 - Also received the Fulbright Critical Language Enhancement Award: A 3-month intensive language immersion project prior to starting the Study/Research award. Completed with Advanced speaking level.

Undergraduate Research

- **Characterizing the Structure of the Martian Bow Shock** 2016 - 2019
 Reference: Associate Professor Jasper Halekas, Univ. Iowa
 - Performed statistical study of the shape and pattern of the Martian bow shock with varied solar wind conditions using data from Mars Atmosphere and Volatile Evolution (MAVEN) Mission using IDL to fit both simple curves to data.
- **Water Pollution of White Dwarf Binaries by Exo-comets** Summer 2017
 Reference: Assoc. Prof. Hagai Perets, Technion Israel Institute of Technology Astrophysics dept
 - Wrote script for REBOUND N-body integrator simulation to explore how mass loss in a white dwarf binary could influence the evolution and dynamics of surrounding comet cloud.
 - Additional project focused on recreating the results of Withers et al. (2017) in order to make predictions for potential sources to use radio occultation methods.
- **Search for thermal gyro-synchrotron emission in pre-main sequence and T-Tauri stars** 2016 - 2020
 Reference: Professor Robert Mutel, Univ. of Iowa
 - Analysis of thermal GS emission from pre-main sequence and T-Tauri stars using new minimization fitting methods.
 - Analysis of VLA data of Ultra-Cool Dwarf star 2M0746+20 as a follow-up to previous paper (Lynch et al. 2015).

PUBLICATIONS

Refereed Publications

- [1] **Lipman, D.**, Battersby, C., Walker, D. L., et al. 2025, The Astrophysical Journal, 984, 159, doi: [10.3847/1538-4357/adb5ee](https://doi.org/10.3847/1538-4357/adb5ee)

- [2] Alboslani, D., Battersby, C., Brunker, S. W., et al. 2025, , 169, 213, doi: [10.3847/1538-3881/adba5f3](https://doi.org/10.3847/1538-3881/adba5f3)
- [3] Brunker, S. W., Battersby, C., Alboslani, D., et al. 2025, , 982, L20, doi: [10.3847/2041-8213/adba5e](https://doi.org/10.3847/2041-8213/adba5e)
- [4] Battersby, C., Walker, D. L., Barnes, A., et al. 2025a, The Astrophysical Journal, 984, 157, doi: [10.3847/1538-4357/adb844](https://doi.org/10.3847/1538-4357/adb844)
- [5] Battersby, C., Walker, D. L., Barnes, A., et al. 2025b, The Astrophysical Journal, 984, 156, doi: [10.3847/1538-4357/adb5f0](https://doi.org/10.3847/1538-4357/adb5f0)
- [6] Sofue, Y., Oka, T., Longmore, S. N., et al. 2025, , Publications of the Astronomical Society of Japan, 2025, doi: [10.1093/pasj/psaf072](https://doi.org/10.1093/pasj/psaf072)
- [7] Tress, R. G., Sormani, M. C., Girichidis, P., et al. 2024, , 691, A303, doi: [10.1051/0004-6361/202450035](https://doi.org/10.1051/0004-6361/202450035)
- [8] Walker, D. L., Battersby, C., **Lipman, D.**, et al. 2025, The Astrophysical Journal, 984, 158, doi: [10.3847/1538-4357/adb5ef](https://doi.org/10.3847/1538-4357/adb5ef)

In-Prep Work

- [1] **Lipman, D.**, Battersby, C., Walker, D. L., et al. *3D CMZ V: A new orbital model of our Galaxy's Center, informed by data across the electromagnetic spectrum. Expected submission mid-October.*
 - Bayesian synthesis of line-of-sight positions of CMZ sources and 4-D fitting method. Presents a top-down view of the Milky Way's CMZ.
- [2] **Lipman, D.** et al. *A Survey of Compact, High-Velocity Features Observed in CS (2–1) with ACES. Expected submission late-October.*
 - Identification and cataloging of over 200 high-velocity features in the CMZ using high resolution ALMA data. A large portion of the analysis and writing was completed in a 2-week intensive paper sprint with collaborators.
- [3] **Lipman, D.** et al. (in-prep)
 - Technical paper presenting new MHD Zoom in simulations.

SELECTED TALKS AND CONFERENCES

Sept 2025	Stellar Origins Conference (Vienna, Austria) – Poster
July 2025	New England Star and Planet Formation Meeting (Yale) – Talk
Oct 2024	UConn Women in Mathematics and Applications Seminar – Invited Talk
Apr 2024	EcoTarium Museum of Science and Nature (Boston, MA) – Invited Public Talk
Feb 2024	Kansas University Astronomy Seminar – Invited Talk
Sept 2023	Arepo Development Workshop for ISM simulations – Participant in collaborative development workshops for new simulation modules
Apr 2023	Protostars and Planets VII (Kyoto, Japan) – Poster

SKILLS

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- **Programming Languages:** Python (advanced), C (proficient), IDL (proficient), basic html
 - **Visualization Tools:** Proficient use of DS9, CARTA, glueviz, and 3-D python visualization libraries
 - **Web Technologies & Version Control:** Proficient use of GitHub, BitBucket
 - **Database Systems:** Globus, Harvard Dataverse, IRSA
 - **Specialized Area:** Development of MHD simulations using AREPO code suite, extensive use of hdf5 and binary filetypes
 - **High Performance Computing:** Proficient use of high-performance computing and slurm. Introductory use of docker and singularity container platforms
 - **Personal Skills:** Intermediate speaking/reading level Mandarin Chinese

LEADERSHIP AND OUTREACH EXPERIENCE

- **Astrophysical Journal Peer Reviewer** 2025
- **UConn Astronomy Seminar Committee** 2022 - Present
- **Collaboration journal club** 2022 - Present
 - Lead organizer of virtual monthly journal club between three US-based research groups. Focused on graduate student-lead presentation of timely research papers.
- **Research Mentorship of Undergraduate Students** 2021 – Present
- **Co-Organizer of Astronomy on Tap Storrs** [🌐] 2022 – 2025
- **Co-lead of UConn STARS: Science Technology and Astronomy Recruits** 2022 - 2024
PI: Prof. Cara Battersby (UConn) [🌐]
 - Program for undergraduate students in physics, aimed to recruit and retain students from historically excluded groups in physics.
 - Organized and led undergraduate in the creation and teaching of lesson plans for under-resourced k-12 schools
- **Hartford Public Library Outreach Events** 2022 - 2024
 - Organized and lead 4 after school programs for k-6 aged children

TEACHING EXPERIENCE

- **Graduate Teaching Assistant** Fall 2020 – Fall 2022
University of Connecticut Department of Physics
 - Assisted and led active-learning classroom lectures, discussion, and labs for Physics II for engineering students
 - Taught traditional 3-hour labs for algebra-based Physics I
- **Undergraduate Teaching Assistant** Fall 2017 – Spring 2019
University of Iowa Department of Physics & Astronomy
 - Assisted in active-learning classroom lectures, and led discussion sections for pre-med focused Physics II, as well as Physics I and Physics II for engineering students.
- **Honors Writing Fellow** Fall 2016 – Fall 2017
University of Iowa Writing Center
 - Fellowship position where awardees are assigned to a professor for a semester to peer review midterm and final papers. Met with students individually to discuss drafts.
- **Writing Center Tutor** Fall 2017 – Spring 2019
University of Iowa Writing Center
 - Tutored scheduled appointments with undergraduate and graduate students from any major throughout the university.